MCB137L / MCB237L
Physical Biology of the Cell

Hernan G. Garcia

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Introduction

Biology is being revolutionized by new experimental techniques that have made it possible to quantitatively query the inner workings of molecules, cells, and multicellular organisms in ways that were previously unimaginable. The objective of this course is to respond to this deluge of quantitative data through quantitative models and the use of biological numeracy. The course will explore the description of a broad array of topics from modern biology using the language of physics, mathematics, and computation. One style of thinking we will emphasize imagines the kinds of simple calculations that one can do with a stick in the sand.

We will draw examples from across modern biology, including cell biology, gene regulation, genomics, developmental biology, and neuroscience, to develop theoretical models that make precise, quantitative predictions about biological phenomena. These predictions will be explored through hands-on analysis of experimental data, numerical simulations, and the development of original scientific projects.

Physical biology will be introduced not simply as a collection of quantitative techniques, but as a way of asking biological questions. Throughout the semester, students will learn how to formulate scientific questions, develop quantitative approaches to answer them, and communicate their findings through presentations and discussion.

No previous coding or advanced mathematics experience is required. The course is designed to be broadly accessible.

Educational Philosophy

Science is often taught as a collection of established facts and techniques. Research, however, proceeds very differently. Scientists spend much of their time deciding which questions are worth asking, determining how those questions can be answered, acquiring whatever new tools are needed along the way, and convincing others that their conclusions are justified.

The objective of this course is to provide an authentic experience of this process in the context of physical biology. The lectures introduce the conceptual framework of physical biology. The laboratory provides an opportunity to use those ideas to investigate an original scientific question. Throughout the semester, students will repeatedly refine their questions, analyses, and presentations through discussions with the instructor, the GSI, and their classmates, mirroring the iterative nature of scientific research.

When and where

The course meets twice each week for two hours.

- **Lecture (2 hours), Location TBD.** Development of the core ideas of physical biology through lectures, discussions, and quantitative exercises.
- **Laboratory (2 hours), Location TBD.** Collaborative investigation of an original scientific question in teams of two.

Online Resources:

- **Course Website**
- Piazza

Instructor and GSI

Instructor:

- Hernan Garcia (hggarcia@berkeley.edu)
- Office hours: TBD, 505 Weill Hall and Zoom

GSI:

- Aaron Fultineer (aaron_fultineer@berkeley.edu)
- Office hours: TBD

NOTE: For transparency, rather than emailing Hernan or Aaron, we encourage you to message us through the course's Piazza website about any questions regarding assignments and class logistics.

Course structure

The weekly lecture covers topics such as biological estimation, dynamics, probability, single-cell genomics, diffusion, exploratory dynamics, pattern formation, entropy and statistical mechanics, gene regulation, and biological networks. Because the lectures and laboratory are tightly integrated, portions of the lecture may evolve in response to the scientific questions pursued by the class.

The weekly laboratory serves as a scientific workshop devoted to the investigation of original scientific questions in physical biology. Students will work in teams of two to identify and explore a question that genuinely interests them. We will suggest possible directions while encouraging students to formulate questions of their own, and every project will be refined through regular discussions with the instructor and GSI so that it remains scientifically interesting, quantitatively grounded, and feasible within the semester. Throughout the semester, we will also discuss and present examples of what we call the *Protocol of Science*: how to identify an interesting scientific problem, formulate quantitative hypotheses, develop appropriate models and analyses, interpret evidence, and communicate scientific results.

Although much of the laboratory will be devoted to collaborative project work, portions of selected laboratory meetings may become hands-on tutorials on computational or mathematical techniques that emerge naturally from the needs of the projects. These may include Python programming, numerical simulations, data visualization, statistical inference, mathematical modeling, artificial intelligence tools, or other techniques required to investigate the scientific questions under consideration. Rather than following a predetermined sequence of technical topics, these mini-workshops will be driven by the scientific questions themselves.

Representative scientific questions include:

- What determines the length of a biological structure?
- How can entropy organize intracellular compartments?

- What governs the spread of an epidemic?
- How do exploratory dynamics accelerate biological search?
- How do the forces of evolution dictate the dynamics of populations?

Three weeks of the semester—Weeks 3, 8, and 14—will be devoted to project milestones and presentations, as described below.

Students are encouraged to use all scientific resources available to them, including textbooks, the primary literature, computational tools, and artificial intelligence systems. Expectations regarding their appropriate use are described immediately below.

Weekly homework sets accompany the lecture material. These are intended for practice and self-assessment and will not be graded, but solutions will be provided to aid in self-study.

Use of Artificial Intelligence and Other Scientific Tools

Students are encouraged to use the scientific literature, computational software, and artificial intelligence tools throughout the course. These resources may be used to explore ideas, identify relevant literature, develop or debug code, test mathematical approaches, analyze data, improve figures, and revise scientific writing. The purpose of the course is not to avoid these tools, but to learn how to use them thoughtfully, critically, and effectively as scientists. In the terms of UC Berkeley's Academic Senate guidance, this is a course in which the use of generative AI is permitted and encouraged, but not required. There is no assignment in this course on which these tools are prohibited, and no student is obliged to use them.

The use of such tools does not transfer responsibility for the work. Students remain fully responsible for the accuracy, originality, and scientific integrity of every idea, calculation, piece of code, figure, argument, and conclusion they submit or present. All substantive use of artificial intelligence must be acknowledged in the relevant written submission, including a brief description of how the tool was used. Students must independently verify AI-generated claims, citations, calculations, and code rather than assuming that they are correct.

Every written submission should therefore close with a short acknowledgement that names each system you used, links to it, states what you used it for, and explains how its output made its way into the work you are submitting. You are not required to submit a complete log of your prompts,

but you should be able to produce representative prompts and their outputs if we ask. When text, analysis, or a figure derived from one of these systems appears in a submission, cite it as you would any other source, in the text and in your reference list. In-text: (OpenAI, 2024). In the references: OpenAI. (2024). ChatGPT (May 2 version) [Large language model]. `https://chat.openai.com/chat`.

Be careful about what you type into these systems. They do not guarantee the privacy of what you send them, and anything you submit may be retained or used for training. Do not enter personally identifiable or sensitive information, including student identification numbers, precise addresses or locations, government-issued identification, financial or credit card information, or personal medical information — in general, anything falling under Protection Levels 3 or 4 of Berkeley's data classification standard.

No project in this course requires a paid subscription to any tool. The freely available tiers of current AI systems, together with Python and the campus computing resources described on the course website, are entirely sufficient for every project we will undertake; the Virtual Computing Lab is also available to you. If access to software or computing ever puts you at a disadvantage, tell us early. We will find a solution, and no project will be evaluated on the basis of what its authors could afford.

We will return to the limitations of these tools during our Protocol of Science discussions in the laboratory: how they produce confident errors and fabricated citations, how bias enters what they generate, what happens to the data you feed them, and how to check their claims against something more reliable. Learning to distrust a tool productively is part of learning to use it.

Most importantly, every student must have intellectual ownership of the work produced by their team. Each student should be able to explain the reasoning behind every major methodological choice, reproduce and interpret the central analyses, identify the limitations of the work, and defend the conclusions during presentations and discussion. Submitting material that a student cannot explain or does not understand will be treated as a failure to meet the academic and scientific expectations of the course, regardless of whether that material was produced by another person, by artificial intelligence, or by another computational tool.

This is also how the policy is enforced. We do not use automated AI-detection software; the campus does not make it available to instructors, and we would not trust it if it did. What we rely on instead is conversation — your ability to explain, reproduce, and defend your own work when we ask you about it. A student suspected of violating this policy will be reported

to the Center for Student Conduct and may be required to redo the work in person.

Project Presentations

Instead of traditional examinations, students will complete three project milestones. Each milestone combines written and oral scientific communication. These written and oral components are designed to mirror the scientific publication process, progressing from an initial research proposal, to a manuscript draft, and ultimately to a completed research article.

- **Week 3: Research Proposal**
  - One-page abstract describing the scientific question, motivation, proposed quantitative approach, and expected outcomes.
  - 10-minute proposal presentation followed by approximately 10 minutes of questions.
  - Evaluation will emphasize the importance and clarity of the scientific question, the physical biology motivation, and the feasibility of the proposed approach.
- **Week 8: Progress Report**
  - A 3–5 page manuscript draft written in the style of a scientific paper, including an abstract, introduction, methods, preliminary results, discussion, and references.
  - 10-minute progress presentation followed by approximately 10 minutes of questions.
  - Evaluation will emphasize scientific and technical progress, the quality of the preliminary analysis, the clarity of the written report, the identification of obstacles and next steps, and the incorporation of feedback.
- **Week 14: Research Article and Final Presentation**
  - A polished 6–8 page research article written in the style of a scientific paper, including an abstract, introduction, methods, results, discussion, and references.
  - 10-minute final presentation followed by approximately 10 minutes of questions.

  - Evaluation will emphasize scientific depth, the quality of the quantitative analysis, the strength of the conclusions, the clarity of the written and oral communication, and intellectual ownership demonstrated during discussion.

Submitting Your Work

All written reports and presentation slides must be submitted and will be graded on GradeScope (entry code TBD), as described on the course website. Any other form of submission will not be accepted.

- Written reports are due at the beginning of class one week before the corresponding presentation. This is intended to give you time to incorporate our feedback into the presentation itself.
- Presentation slides are due at the beginning of class on the day of the presentation.
- All code used to produce the results you present needs to be submitted through GradeScope by the slide deadline.
- No late submissions. Time management is key. Start early and make use of office hours and our availability over Piazza.
- It is important to describe your reasoning. Just writing an equation or showing a plot does not constitute a satisfactory scientific argument.
- All plots need to have labeled axes.
- When reporting on magnitudes, a number of significant digits that is concomitant with the calculation being carried out should be used. Failure to do so will be interpreted as lack of understanding of the basic principles of estimation.

Concretely, this means the proposal abstract is due in Week 2, the progress report in Week 7, and the research article in Week 13.

Expectations for MCB237L

Projects may be completed by teams consisting of undergraduate students, graduate students, or a combination of both. Students enrolled in MCB237L will be expected to contribute at a level appropriate for graduate study,

including greater engagement with the primary literature, additional quantitative or computational depth where appropriate, and leadership in the conception, execution, and communication of the project.

Tentative Lecture Syllabus

- Week 1: A Feeling for the Numbers in Biology
  - Street-Fighting Mathematics: Order-of-magnitude estimates as a tool for discovery in the living world.
  - What sets the scale of X? Exploring the limits to bacterial growth.
- Week 2: Stuff(t)
  - Time evolution in biology.
  - Update rules.
  - Control of bacterial growth and size.
  - Transcription of the constitutive promoter.
- Week 3: Research Proposal Presentations
- Week 4: Null Hypotheses in Biology: The Poisson Distribution.
  - The Great Probability Distributions of Biology.
  - The Poisson distribution: Bombs over London, constitutive promoters and sequencing the human genome.
- Week 5: The Synthetic Transcriptome
  - The single-cell sequencing revolution.
  - The physics of single-cell sequencing.
  - Dimensionality reduction for the transcriptome.
- Week 6: Null Hypotheses in Biology: The Exponential Distribution
  - Waiting times for photobleaching and ion channel dynamics.
  - The constitutive promoter revisited.
- Week 7: Diffusion as Biology's Null Hypothesis for Dynamics
  - Diffusion and axonal transport.

  - The binomial distribution: Coin flips, carboxysome partitioning and calibrating fluorescent protein counts.
  - FRAP: Measuring diffusion using photobleaching.
  - A universal diffusion speed limit for enzyme catalysis and other reactions.

- Week 8: Progress Report Presentations
- Week 9: Exploratory Dynamics
  - A new kind of update rule: purposeful dynamics.
  - The time for transcription factors to find their targets.
  - How do microtubules find chromosomes.
- Week 10: Biological Patterns in Space and Time
  - Morphogen gradients and the French flag model.
  - Turing patterns and digit formation.
- Week 11: Entropy Rules
  - Entropy maximization and the mathematics of superlatives.
  - Entropic forces: A DNA entropic spring.
  - Phase transitions in biology.
- Week 12: The Boltzmann Distribution and Statistical Mechanics
  - Ion channels and two-state systems.
  - The ubiquitous nature of binding problems in biology.
  - Regulatory biology: simple repression in the *lac* operon.
  - Whence cometh the allosterome?
- Week 13: Biological Networks
  - A life-or-death decision: The Lambda switch.
  - Making Waddington real.
  - Vertebrate development and oscillations.
- Week 14: Research Article and Final Presentations

Participation and attendance

Because the course revolves around collaborative scientific work and iterative feedback, attendance and active participation are essential. Students are expected to attend every lecture and laboratory meeting. Repeated unexcused absences will negatively affect the participation component of the final grade. Students who must miss a class meeting or project milestone because of illness, disability accommodations, religious observance, or another serious circumstance should contact the instructor and GSI by email as early as possible so that an appropriate alternative can be arranged. Personal circumstances of this kind are an explicit exception to our general preference for Piazza.

Regular interaction with the instructor and GSI is a central part of the course. Students are therefore expected to be able to attend at least one of the two weekly office hours on a regular basis. If a student's schedule makes this impossible, they should carefully consider whether this course is compatible with their schedule, as these discussions are intended to play an important role in the development of their scientific projects.

Please also notify us in writing by the second week of the semester about any known or potential academic or extracurricular conflicts. We will try our best to help you with making accommodations, but cannot promise them in all cases. Finally, if you do miss a class, it is your responsibility to get notes from one of your classmates; you cannot expect the instructor or GSI to redo the lecture during office hours.

Grading

- Class participation: 15%
- Research proposal (abstract and presentation): 15%
- Progress presentation: 10%
- Progress report: 15%
- Final presentation: 20%
- Research article: 25%

We do not grade on a curve. The grading scale we will use is shown below.

Letter	Percentage
A	94–100
A-	90–93
B+	87–89
B	84–86
B-	80–83
C+	77–79
C	74–76
C-	70–73
D+	67–69
D	64–66
D-	60–63
F	Below 60

Course Help

You're not alone in this course; the instructor and GSI are here to support you as you learn the material. We are committed to helping you succeed and get the most out of this course. It's expected that some aspects of this course will take time to grasp, and the best way to grasp challenging material is to ask questions. To ask a question, use Piazza, or attend office hours. **In fact, we strongly encourage you to make time in your schedule to attend at least one of the two office hours we offer.** The instructor and GSI will monitor this discussion forum, but you should also feel free to answer questions posted by other students.

Students with Disabilities

If you require course accommodations due to a physical, emotional, or learning disability, contact UC Berkeley's Disabled Students' Program (DSP). Notify the instructor and GSI by email — rather than through Piazza — of the accommodations you would like to use, so that the matter remains private. You must have a Letter of Accommodation on file with UC Berkeley to have accommodations made in the course.

Academic Integrity

You're a member of an academic community at one of the world's leading research universities. Berkeley creates knowledge that has a lasting impact

in the world of ideas and on the lives of others; such knowledge can come from an undergraduate paper as well as the lab of an internationally known professor. One of the most important values of an academic community is the balance between the free flow of ideas and the respect for the intellectual property of others. Scholars and students always use proper citations in papers; professors may not circulate or publish student papers without the writer's permission; and students may not circulate or post materials (handouts, exams, syllabi—any class materials) from their classes without the written permission of the instructor.

Any test, paper or report submitted by you and that bears your name is presumed to be your own original work that has not previously been submitted for credit in another course unless you obtain prior written approval to do so from your instructor. In all of your assignments, including your homework or drafts of papers, you may use words or ideas written by other individuals in publications, websites, or other sources, but only with proper attribution. If you're unclear about the expectations for completing an assignment or taking a test or examination, be sure to seek clarification from your instructor or GSI beforehand. For additional information on plagiarism and how to avoid it, read the UC Berkeley Library Citation Page, Plagiarism Section.

The permitted use of artificial intelligence and other computational tools in this course is described above. Such use is consistent with academic integrity only when it is appropriately acknowledged, critically evaluated, and accompanied by full intellectual ownership of the submitted work.

As a member of the campus community, you're expected to demonstrate integrity in all of your academic endeavors and will be evaluated on your own merits. The consequences of cheating and academic dishonesty—including a formal discipline file, possible loss of future internship, scholarship, or employment opportunities, and denial of admission to graduate school—are simply not worth it. Read more about Berkeley's Honor Code.

Suggested reading

- Phillips, R. *et al.* (2012). Physical Biology of the Cell, 2nd Edition. Garland Science.
  - Our course will be loosely based on this textbook. Many of the assigned problems will come from it. Note that we are actively working on the 3rd edition and I might hand out some chapters as we visit their topics in lecture.

- Kinder, J. and Nelson P. (2018). A Student's Guide to Python for Physical Modeling: Updated Edition. Princeton University Press.
  - This book is a great introduction to programming in Python. It could be useful reference material throughout the semester.
- Alberts, B. *et al.* (2014). Molecular Biology of the Cell. W. W. Norton & Company.
  - This book will be particularly useful for those needing a refresher of biology.
- Milo, R. and Phillips, R. (2015). Cell Biology by the Numbers. Garland Science.
  - This book, together with its companion BioNumbers website, has become the reference source for biological numeracy. The book can also be downloaded from `book.bionumbers.org`.
- Mahajan, S. (2010). Street-Fighting Mathematics: The Art of Educated Guessing and Opportunistic Problem Solving. MIT Press.
- Weinstein, L. and Adam, J.A. (2008). Guesstimation: Solving the World's Problems on the Back of a Cocktail Napkin. Princeton University Press.
  - These two books are fantastic resources for those wanting to learn more about estimation writ large.